SPLITTING FIELDS OF A CENTRAL SIMPLE ALGEBRA AND SIMILAR ALGEBRAS OF THE SAME DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both directions; every identification is either displayed or justified by a named preceding result, and the closing sentence 'so we win anyway' follows the displayed dimension computation.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Self centralizing subfields. Same chapter, Lemma lemma-self-centralizing-subfield. Supplies that the image of the field in the opposite algebra is its own centralizer, which identifies the tensor product with the centralizer of the ground field inside an endomorphism algebra.
- The commutant and the Brauer group. Same chapter, Lemma lemma-when-tensor-is-equal. Supplies that the class of the commutant is the negative of the class of the algebra in the Brauer group, which gives similarity of the two algebras.
- The dimension formula for the centralizer. Same chapter, Theorem theorem-centralizer. Supplies the identity relating the degrees of the algebra and of its commutant, from which the degree of the chosen algebra is computed.

Source reference. Original lemma 074V: sources/stacks-brauer/pinned-original-source.tex, source lines 800–800. Navigation pointer invoked at proof line 828; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074U: sources/stacks-brauer/pinned-original-source.tex, source lines 784–784. Navigation pointer invoked at proof line 878;

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Source reference. Original theorem 074T: sources/stacks-brauer/pinned-original-source.tex, source lines 746–746. Navigation pointer invoked at proof line 793; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Theorem 0.1. *Let A be a finite central simple k-algebra. Let k'/k be a finite field extension. The following are equivalent*

(1) *k' splits A, and*
(2) *there exists a finite central simple algebra B similar to A such that $k' \subset B$ and $[B : k] = [k' : k]^2$.*

Proof. Assume (2). It suffices to show that $B \otimes_k k'$ is a matrix algebra. We know that $B \otimes_k B^{op} \cong \mathrm{End}_k(B)$. Since k' is the centralizer of k' in B^{op} by Lemma 074V we see that $B \otimes_k k'$ is the centralizer of $k \otimes k'$ in $B \otimes_k B^{op} = \mathrm{End}_k(B)$. Of course this centralizer is just $\mathrm{End}_{k'}(B)$ where we view B as a k' vector space via the embedding $k' \to B$. Thus the result.

Assume (1). This means that we have an isomorphism $A \otimes_k k' \cong \mathrm{End}_{k'}(V)$ for some k'-vector space V. Let B be the commutant of A in $\mathrm{End}_k(V)$. Note that k' sits in B. By Lemma 074U the classes of A and B add up to zero in $\mathrm{Br}(k)$. From the dimension formula in Theorem 074T we see that

$$[B : k][A : k] = \dim_k(V)^2 = [k' : k]^2 \dim_{k'}(V)^2 = [k' : k]^2[A : k].$$

Hence $[B : k] = [k' : k]^2$. Thus we have proved the result for the opposite to the Brauer class of A. However, k' splits the Brauer class of A if and only if it splits the Brauer class of the opposite algebra, so we win anyway. □